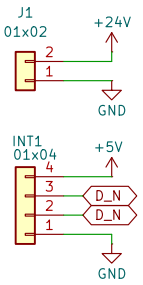
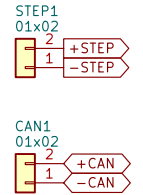


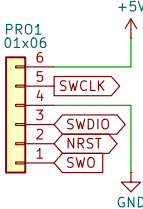
Power & Interconnect



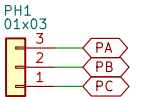
Control Interfaces



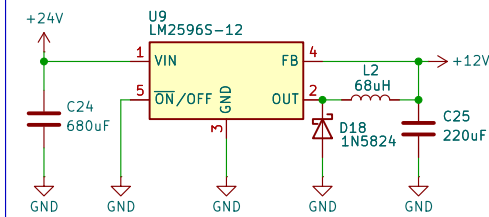
Programming Interface



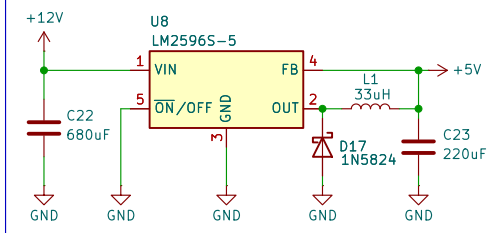
3-Phase Interface



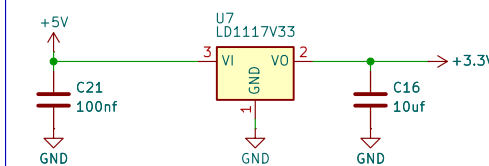
Step down regulator (24V to 12V)



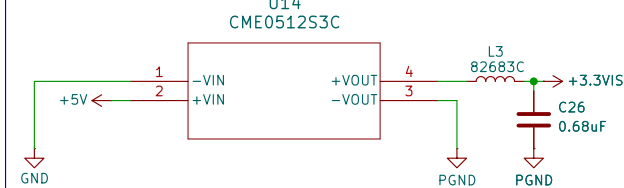
Step down regulator (12V to 5V)



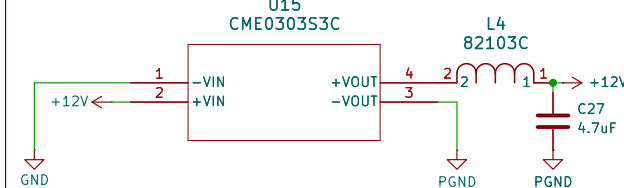
Low dropout regulator (5V to 3.3V)



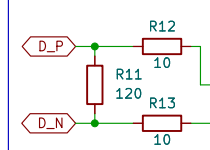
Isolated 3.3V Supply



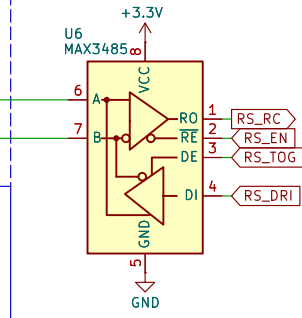
Isolated 12V Supply



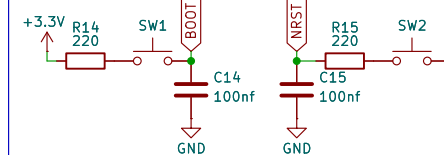
RS-485/RS-422 Terminations



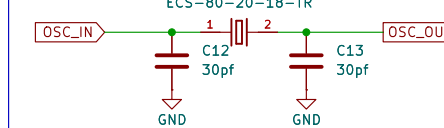
RS-485/RS-422 Transceiver



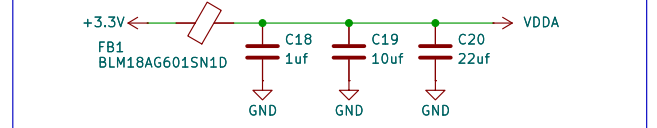
Boot / Reset



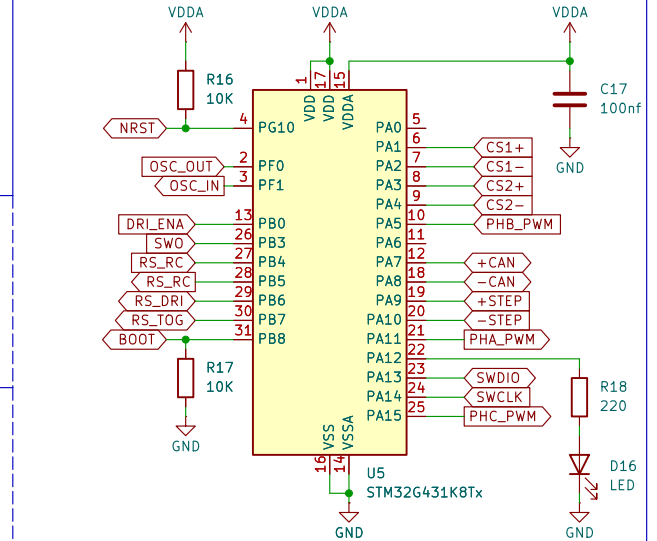
8 Mhz Oscillator



VDDA Decoupling



Controller (170 Mhz)



WARNING:

For current sensing, use either the '1 mΩ', '3 mΩ', or '10 mΩ' shunt resistor. Do not use more than a single path, or the driver will not function correctly.

drivers

File: Drivers.kicad_sch	current
File: current.kicad_sch	

OpenLSM

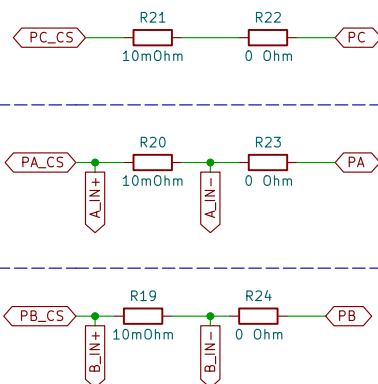
Sheet: /
File: main.kicad_sch

Title: Bridge Driver

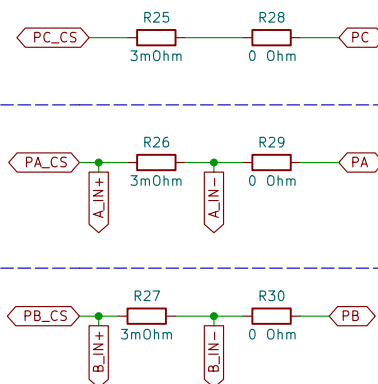
Size: A4 Date: 2026-08-03
KiCad E.D.A. 10.0.0

Rev: 0
Id: 1/3

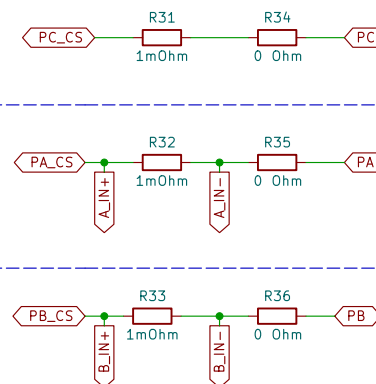
Phase Connections (upto 5.0 A (PEAK))



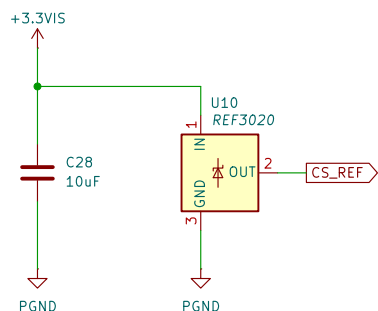
Phase Connections (upto 16.6 A (PEAK))



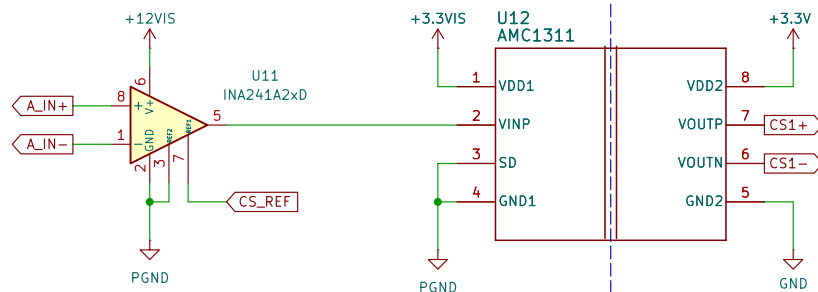
Phase Connections (upto 30 A (PEAK))



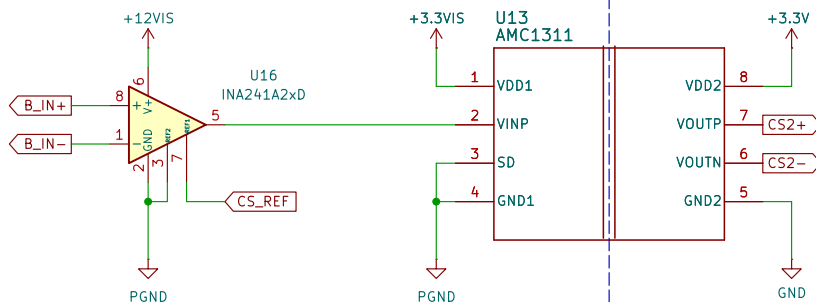
Voltage Follower Reference (+2 V)



Current Sensor (Phase A)



Current Sensor (Phase B)



OpenLSM

Sheet: /current/
File: current.kicad_sch

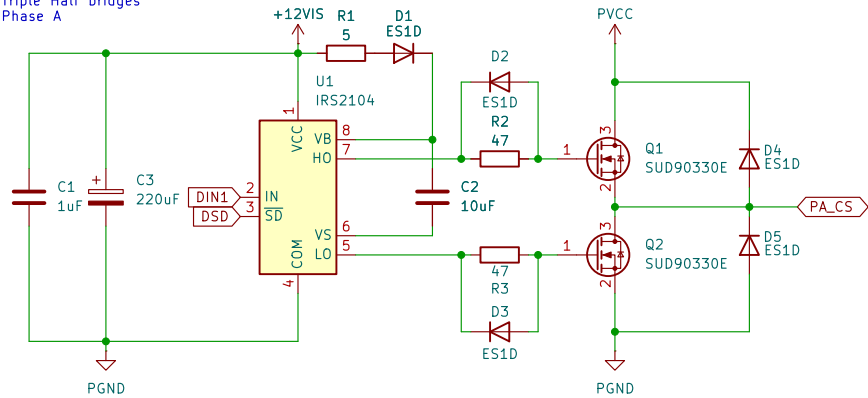
Title: Current Sensors & Isolation

Size: A4
KiCad E.D.A. 10.0.0

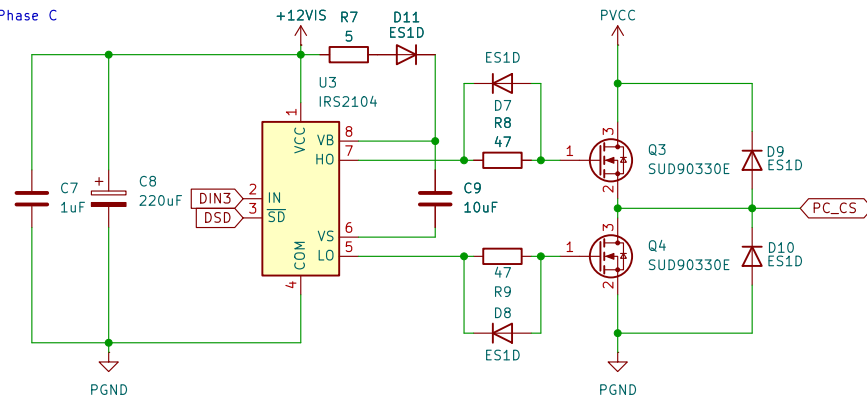
Date:

Rev: 0
Id: 2/3

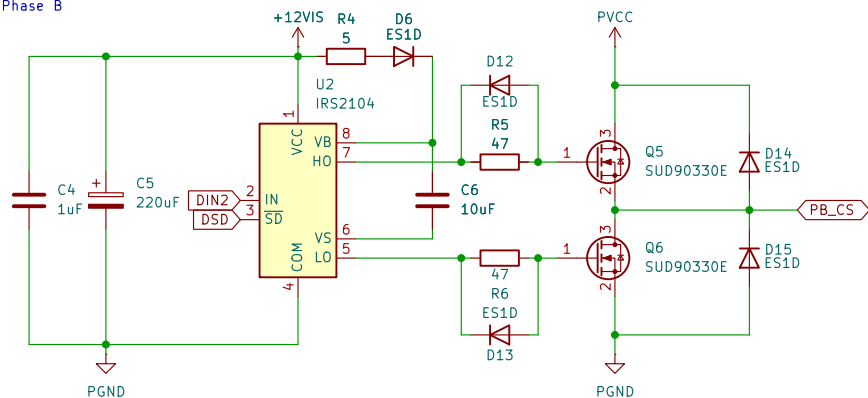
Triple Half bridges
Phase A



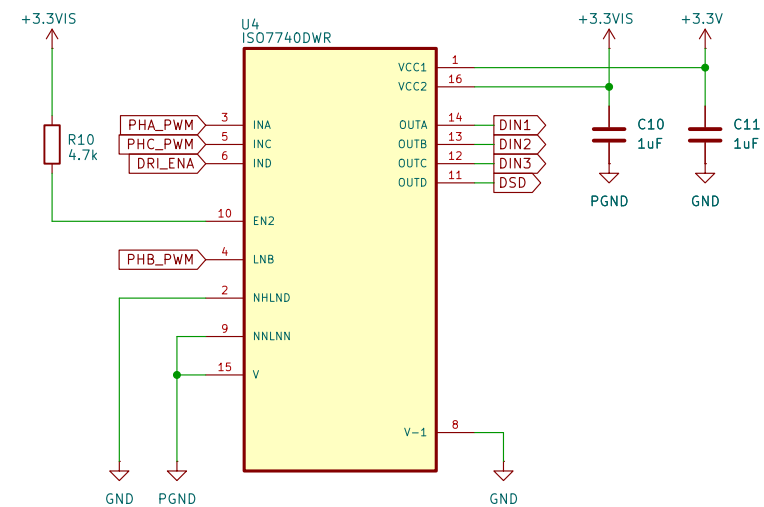
Phase C



Phase B



Isolated Gate Driver Interface



OpenLSM

Sheet: /drivers/
File: Drivers.kicad_sch

Title: Phase Drivers & Isolation

Size: A4
KiCad E.D.A. 10.0.0

Date:

Rev: 0
Id: 3/3